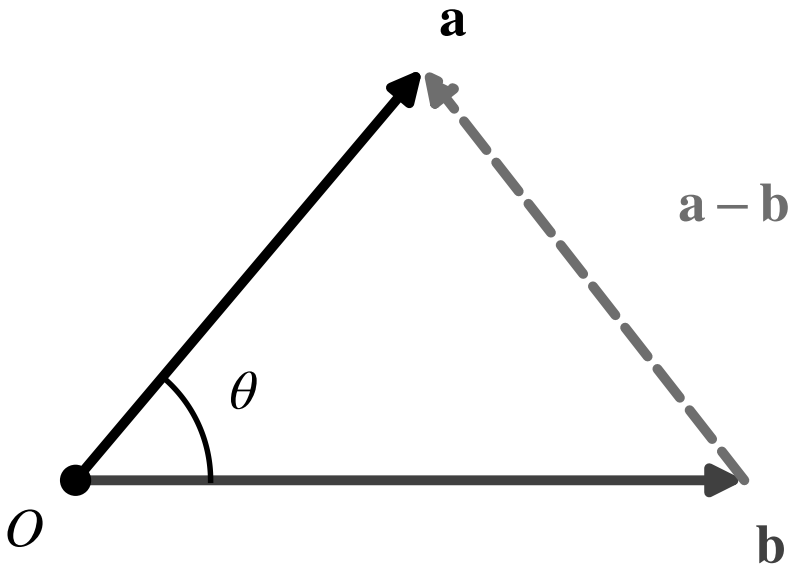


$$\|\mathbf{a}-\mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\,\|\mathbf{a}\|\,\|\mathbf{b}\|\cos\theta$$

$$(\mathbf{a}-\mathbf{b})\cdot(\mathbf{a}-\mathbf{b}) = \mathbf{a}\cdot\mathbf{a} - 2\,\mathbf{a}\cdot\mathbf{b} + \mathbf{b}\cdot\mathbf{b}$$

消去 $\|\mathbf{a}\|^2 + \|\mathbf{b}\|^2$ 后：$\mathbf{a}\cdot\mathbf{b} = \|\mathbf{a}\|\,\|\mathbf{b}\|\cos\theta$